



Detection of linezolid resistance *cfr* gene among MRSA isolates

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ABSTRACT

Background: Linezolid (Oxazolidinones) is commonly used against a variety of Gram-positive infections, especially methicillin-resistant *Staphylococcus aureus* (MRSA). The emerging resistance to linezolid curtail the treatment of infections caused by MRSA and other Gram-positive bacteria. Presence of *cfr* gene plays a crucial role in Linezolid resistance.

Objective: Present study was aimed to detect *cfr* gene among clinical MRSA isolates.

Materials and methods: The suspected *Staphylococcus aureus* isolates were processed through Kirby Bauer disc diffusion methods for the confirmation of MRSA strains. Phenotypic Linezolid resistance was determined through broth micro-dilution method. The plasmid and DNA of Linezolid resistant isolates were subjected to molecular characterization for the presence of *cfr* gene.

Results: Among 100 *Staphylococcus aureus* isolates, 85 of them were confirmed as MRSA isolates. Categorically, 65% MRSA isolates were sensitive to linezolid with MIC lower than 8 µg/ml, whereas, 35% of them were resistant to linezolid having MIC greater than 8 µg/ml. MIC level of 128 µg/ml was observed among 3.5% of the resistant isolates. Similarly, MIC level of 64 µg/ml, 32 µg/ml, 16 µg/ml and 8 µg/ml were noted for 3.5%, 4.7%, 8.2% and 15.3% isolates respectively. Linezolid resistance *cfr* gene was detected only in 9.4% of the resistant isolates.

Conclusion: Multi drug resistance among MRSA isolates is keenly attributed to the presence of *cfr* gene as evident in the present study, and horizontal dissemination of *cfr* gene among MRSA strains is accredited to *cfr*-carrying transposons and plasmids.

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Introduction

Methicillin resistant *Staphylococcus aureus* first appeared in UK in 1961. Since then it has disseminated on a global scale and now considered amongst the most antibiotic resistant bug [4]. In Pakistan, first documented reports on MRSA was published in 1989.

Abbreviations: *cfr*, chloramphenicol–florfenicol resistance; MIC, minimum inhibitory concentration; MRSA, Methicillin-resistant *Staphylococcus aureus*

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Clinical outcomes of MRSA infections ranges from mild to severe skin infection, bacteremia and eye infections, and meningitis [5].

In treatment, Linezolid is the most suitable and approved drug of final resort for vancomycin resistant MRSA strains.

Linezolid is a bacteriostatic drug that interferes with microbial protein synthesis. It acts by binding to the adenine region located in the peptidyl transferase center (PTC) at domain V of the 23 S rRNA of the 50 S ribosomal subunit. Broth macro dilution method is preferentially used for determining MIC of linezolid.

The current CLSI breakpoints suggests that linezolid mic (≤ 4 µg/ml) is regarded as susceptible while ≥ 8 µg/ml is considered as resistant.

A significant increase in the rates of acquired antimicrobial resistance (AMR) in *Staphylococcus aureus* has occurred in recent decades, either through horizontal gene transfer of resistance

determinants from mobile genetic elements (MGEs) or through mutations that alter gene expression or binding sites in native genes. The most commonly observed mechanism is the acquiring of the chloramphenicol-florfenicol resistance (*cfr*) gene through vertical and horizontal gene transfer. The gene was detected on the plasmid pSCFS1 which is 16.5-kb and multi-drug resistant. The enzyme *cfr* methyltransferase encoded by *cfr* also contributes to resistance. It acts by targeting the adenine residue at position 2503 in the 23 S rRNA, thereby preventing the binding of five antimicrobial drug classes, including Pleuromutins, and Streptogramin A, Oxazolidinones, Phenicol, Lincosamides (LaMarre et al., 2013). Besides the above means of resistance against linezolid, chromosomal mutations in the genes that encode the 50 S ribosomal proteins L3 (*rplC*), L4 (*rplD*), and L22 (*rplV*) mediate resistance to linezolid [6].

In addition to *cfr*, a novel ABC transporter plasmid known as *optrA* also mediates linezolid resistance along with phenicol. A study in *Staphylococcus epidermis* reported a combined presence of *optrA* and *cfr*. This can result in increasing the MIC of linezolid by 2 fold [2].

The first *cfr* positive strains in clinical *Staphylococcal* samples were detected in 2005 from Ireland and Colombia. This Irish isolate M05/0060 possessed the *cfr* gene on pSCFS7 plasmid, which shared a similarity to plasmid pSCFS3 obtained from a veterinary isolate. The CM05 possessed chromosomally located the *cfr* gene and the IS21–558 element [7].

Therefore, present study was aimed to detect *cfr* gene among clinical MRSA isolates.

Materials and methods

Study design and settings

Present study was conducted in the Microbiology Laboratory, Khyber Medical University (KMU) Peshawar, Pakistan from September 2018 to July 2019.

Sample collection and processing

The samples were collected from the Pathology Departments among the tertiary care hospitals in Peshawar. Based on the origin, samples were obtained from blood, wounds (a breach in the skin, which can lead to infections), Pus (wound drainage), lesions (a region of tissue that has suffered damage through injury or disease), throat and nasal swabs following standard microbiological protocol. Samples were immediately transported to Microbiology Laboratory, KMU. The isolates were identified as *S. aureus* using colony morphology via growing on Blood agar base and Mannitol Salt Agar, Gram staining, catalase and coagulase tests.

Methicillin resistance screening and antimicrobial susceptibility testing (AST)

Methicillin resistance *Staphylococcus aureus* strains were detected using modified Kirby Bauer disk diffusion method according to CLSI guideline 2017 (Ref), where cefoxitin disks were applied in combination of other antibiotic disks. Precisely, Lawn culture was prepared using Mueller Hinton agar medium according to CLSI guideline 2017 (Ref). The antibiotic disks used were Cefoxitin 30 µg, Daptomycin 30 µg, Gentamicin 10 µg, erythromycin 15 µg, Doxycycline 5 µg, Ciprofloxacin 5 µg, Methicillin 30 µg, Clindamycin 2 µg, Trimethoprim 5 µg, Chloramphenicol 30 µg, Rifampin 5 µg antibiotics disks were used by placing it at a distance of 20–25 millimeters apart from each other).

The methicillin resistance was confirmed by Cefoxitin disc. All plates were incubated at 35 °C for 24 h. For quality control *S. aureus* ATCC 25293 was used.

Determination of MIC for linezolid using broth micro-dilution

To determine the MIC for linezolid, antibiotic powder (sigma Aldrich, Italy) dissolved in water was used for Broth Micro-dilution using Mueller Hinton broth (MHB). The results were interpreted as per CLSI Guidelines. The concentration used, ranged from 0.25, 0.5, 1, 2, 4, 8, 16, 32, 64, 128 µg/ml. The protocol for the calculating the concentration has been described previously [8] with a slight modification by using 96 well micro-titer plate. The columns 1–10 contained the Concentration of linezolid while the rows had the strains of MRSA for which the MIC was to be determined. For negative control column 11 (from right to left) nutrient broth (negative control) was used whereas, column 12 (positive control) *S. aureus* ATCC 29212 was used.

Molecular detection of *cfr* and *mecA* gene

Genomic and Plasmid DNA were extracted using commercial Kits (thermo Scientific, USA) and Novel Gene Plasmid Mini-prep Kit were used according to the manufacturer protocol. The Primers used has been Described previously for *mecA* [3] and *cfr* (Macrogen, South Korea). PCR amplification condition for both has Denaturation temperature of 94 °C for 30 s, Annealing temperature of 56.2 °C for *mecA* and 57.6 °C for *cfr* gene. The amplified PCR product was visualized with an ultra violet trans-illuminator and the picture was taken with the help of gel imaging document system. The positive control strain for CFR denoted as *Enterococcus faecalis* 603 50427 X [1] was used.

Results

Bacterial isolates

A total of 100 suspected non-duplicated MRSA isolates were collected from tertiary care hospitals in Peshawar. Higher proportion of MRSA strains were obtained from both Rehman Medical Institute (RMI) and Hayatabad Medical Complex (HMC) as shown in Table 1. Based on the origin of samples, MRSA strains were isolated from blood with 42% among all other specimens shown in Table 2.

The MIC of linezolid

Among 85 of isolates, 35% were linezolid resistant while 65% were observed to be sensitive according to CLSI guidelines where values of 8 µg/ml and above are regarded as resistant level for linezolid as shown in (Table 3) also evident by Fig. 1. The remaining 15 suspected isolates were not confirmed as MRSA as they were unable to show resistance against cefoxitin and penicillin, probably by lacking *mecA* gene.

Out of 85 MRSA isolates 55 isolates were found linezolid sensitive while 30 isolates show MIC greater than 4.

Specifically, 3 of the isolates have MIC value of 0.5 µg/ml, while 24 isolates have MIC value of 1 µg/ml, 20 had MIC value of 2 µg/ml. Furthermore, 4 MRSA isolates had MIC 4 µg/ml, 13 have 8 µg/ml, 7 have MIC 16 µg/ml 4 had isolates of 32 µg/ml, 3 isolates had 64 µg/ml

Table 1
Total number of isolates collected from different Hospitals.

	Hospital Name	Suspected MRSA	Confirmed MRSA
1	Rehman Medical Institute (RMI)	30	27
2	Combined Military hospital	25	21
3	Hayatabad Medical Complex (HMC)	30	27
4	Khyber Teaching Hospital	15	10
	Total	100	85

Table 2
MRSA isolated from different clinical samples.

S.No	Source of Specimen	Number of Isolates	Percentage
1	Blood	42	42%
2	Wounds	28	28%
3	Pus	19	19%
4	Nares	8	8%
5	Lesions	3	3%
	Total	100	100%

Table 3
MIC concentration results against linezolid.

Mic concentration in ug/ul	No of isolates positive
128	3
64	3
32	4
16	7
8	13
4	4
2	20
1	24
0.5	3
0.25	4

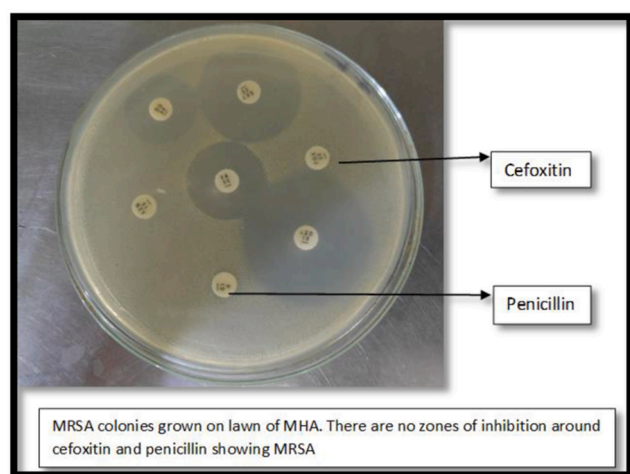


Fig. 1. MRSA confirmation on MH.

and 3 of them had 128 µg/ml respectively. According to CLSI guide lines break point (MIC) of linezolid is 8 µg/ml. The results of this study concludes that 65% of MRSA isolates exhibited sensitivity to linezolid. On the other hand, 35% isolates were found linezolid resistant.

Molecular analysis

Amongst the MRSA resistant strains, the *cfr* gene was detected in 9.41% of the total isolates as shown in Fig. 2. Similarly, all of the phenotypic resistant strains to cefoxitin were confirmed as MRSA by the presence of *mecA* gene shown in Fig. 3.

Discussion

Pathogenic bacteria are gaining resistance to antibiotics used to treat humans and animals, which is causing increasing concern. A wide and chemically diverse category of antibiotics target the ribosome, which is a primary site of antibiotic activity in the bacterial cell. Antibiotic binding sites are concentrated in ribosome functional centers such as the 30 S subunit's decoding center, the peptidyl transferase center (PTC), the GTPase center, and the peptide escape tunnel on the 50 S subunit. Mutation or methylation can cause changes to the ribosomal antibiotic binding sites.

Linezolid has been used to treat a wide range of infections caused by Gram Vancomycin-resistant Enterococcus strains and Staphylococcus aureus for many years (Paiva and Eggimann, 2017).

There are numerous mechanisms that impart resistance. To exercise its antibacterial effect, linezolid binds to the 50 S ribosomal subunit of the 23 S ribosomal RNA (rRNA) and prevents proper protein synthesis. G2576T, T2500A, and G2447T are all 23 S rRNA gene variants that can cause linezolid resistance. Linezolid resistance is also linked to mutations in PTC's ribosomal proteins L3, L4, and L22.

cfr methyltransferase, which is found on the chromosome or resistant plasmids, causes substantial linezolid resistance. *cfr* gene positive MRSA bacteria showed 4–16-fold higher linezolid MICs than *cfr* gene negative germs, according to studies, and *cfr*-positive MRSA isolates had greater antibiotic resistance rates. *Optra*, *fexA*, and *ermC* were found in *cfr*-positive MRSA isolates. Horizontal transmission was linked to the co-dissemination of *cfr* with *fexA* and *ermC*.

For many years, linezolid has been effectively used against wide array of infections caused by both Gram-positive bacteria

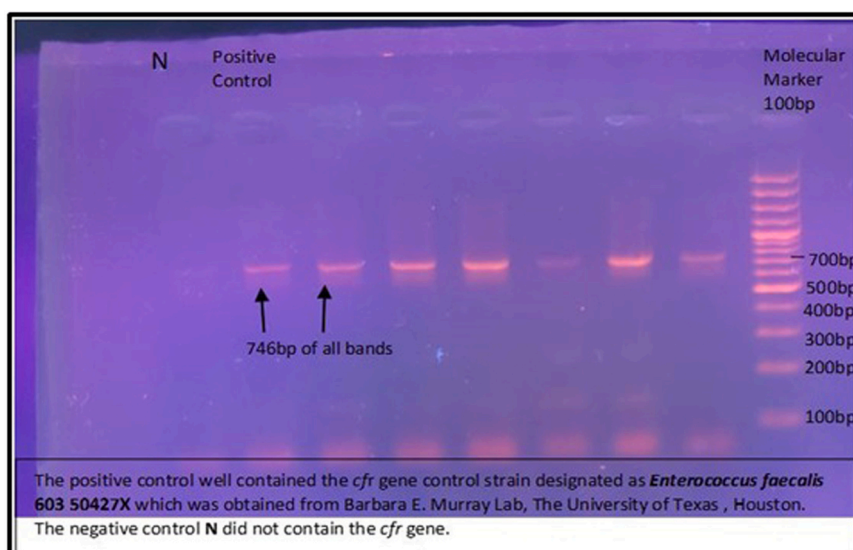


Fig. 2. A representative gel electrophoresis image showing PCR product of *cfr* gene at 746 bp in clinical isolates (The molecular size marker used is a 1 kb ladder).

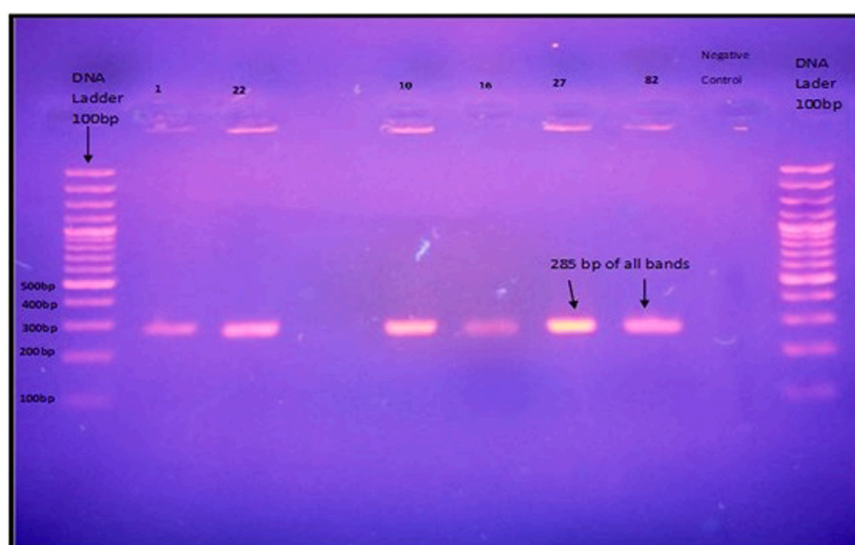


Fig. 3. A representative gel electrophoresis showing PCR product of *mecA* gene at 285 bp in clinical isolates. (The molecular size marker used is a 1 kb ladder).

particularly Vancomycin-resistant *Enterococcus* strains and Vancomycin-intermediate *Staphylococcus aureus*. Many mechanisms confer bacterial resistance to linezolid. The most commonly observed mechanism is the acquiring of the *cfr* gene through vertical and horizontal gene transfer (LaMarre et al., 2013). Amongst clinical MRSA isolates, Toh et al., first detected *cfr* gene mediating resistance against linezolid, in Columbia, USA in pneumonia patient. The patient was hospitalized and administered with linezolid for only two doses. The MRSA (CMO5) isolate had the *cfr* gene chromosomally and it lacked the gene on Plasmid (Toh et al., 2007). Similarly, in our study the *cfr* gene was detected on chromosomal DNA among 9.4% (8) isolates. Another report of *cfr* positive *Staphylococcus aureus* which predates the MRSA CMO5 by five months also published their report of a female patient with Cystic fibrosis. The isolate was denoted as *Staphylococcus aureus* 1128105. The pulsed-field gel electrophoresis (PFGE) results of the isolate revealed a unique molecular typing results in which *cfr* was detected on a plasmid and it also had the mutation in 23 S rRNA sequence that rendered the isolate as resistant strain (Locke et al., 2014).

In Spain, Ruiz de Gopegui et al. (2012), describe the presence of plasmid bearing *cfr* MRSA isolate of pandemic Pantone Valentin Leucocidin (PVL). Unlikely in our study no plasmid bearing *cfr* MRSA isolate was detected. The possible reason for this variation could be the existence of a variety of mechanisms attributed to the resistance against linezolid.

Another documented study of *cfr* gene from Monsoura University Hospital in Egypt. The study accounted for 197 isolates, amongst which 179 were linezolid resistant. Most of the isolates were obtained from blood sample with 41% followed by wounds and pus specimens. Similarly, in our study 42% (42) MRSA isolates obtained from blood followed by 28% (28) wounds, 19% pus, 8% Nares and 3% lesions respectively. The linezolid resistance was checked by agar dilution method. The MIC concentration for the study ranged between 8 µg/ml to 32 µg/ml (El-Kady et al., 2017). In comparison to this study higher MIC values of 128 µg/ml and 64 µg/ml were noted in our study.

Possible reason for this much high MIC values could be due to the multiple resistance mechanism adopted by MRSA against linezolid.

In another study a higher proportion of MRSA were isolated (280) from blood by Ullah et al. (2016) in Peshawar, where 48.7% strains were isolated from blood. Similarly, in our study 42% of the isolates were recorded from blood samples. The difference in the study lies due to higher number of samples in his study and the isolates were

collected from only one tertiary hospital. In contrast, our samples were obtained from different hospitals. In addition, their study did not include the frequency of *cfr* gene. Kehenberg and Schwarz confirm the gene *cfr* in only 3% of their isolates while our study confirms 9.41%, which is significantly higher (El-Kady et al., 2017).

Documented report from [9] in China reports the dissemination of *cfr* gene in animal origin *Staphylococcal* isolates. The prevalence was noted for 18.6% of the isolates. The presence of *cfr* gene in our region can be supported by the fact that there is selective pressure of antibiotics due to misuse in both clinical and veterinary environments.

The emergence of resistant gene such as *cfr* should be a matter of immediate concern as this can spread rapidly to other microbes especially for Gram-positive bacterial species where linezolid is the drug mostly preferred when other antibiotics fail to work. This resistance if spread will leave us with very limited option to treat infections especially in cases of MRSA.

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Declarations of interest

None.

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